When Is Sharing Free?

An Exact Criterion for Zero Additional Randomness

Abstract

For a prescribed finite joint distribution, we characterize when additional recipients require no additional seed entropy. The result concerns randomness, not physical implementation cost. No claim of originality is made.

Let $X_1, \dots, X_N$ be finite-valued random variables with a prescribed joint law. An exact shared-seed realization consists of a discrete random variable S and deterministic maps f_i such that

$$(f_1(S), \dots, f_N(S)) \stackrel{d}{=} (X_1, \dots, X_N).$$

All entropies are measured in bits. Define

$$C(X_1, \dots, X_N) := \inf H(S),$$

where the infimum ranges over all exact realizations. The prescribed joint law, including its correlations, is part of the specification.

Theorem 1 (Exact sharing criterion)**.** *The minimum is attained and satisfies*

$$C(X_1, \dots, X_N) = H(X_1, \dots, X_N).$$

Consequently,

$$\boxed{C(X_1, \dots, X_N) - C(X_1) = H(X_2, \dots, X_N \mid X_1).}$$

In particular, the additional seed cost is zero if and only if there are deterministic maps $g_2, \dots, g_N$ such that

$$X_i = g_i(X_1) \quad \textit{almost surely}, \qquad i = 2, \dots, N.$$

Proof. For every exact realization, the output vector is a deterministic function of S. Hence

$$H(X_1, \dots, X_N) \le H(S).$$

Equality is attained by choosing the output vector itself as the seed and taking the coordinate projections as decoders. The displayed difference now follows from the entropy chain rule.

For finite-valued variables, conditional entropy vanishes precisely when every conditional distribution on an event $\{X_1 = x\}$ of positive probability is a point mass. Choosing that point as $(g_2(x), \dots, g_N(x))$ proves the equivalence. □

Corollary 1 (Replication versus independent experience)**.** *If $X_1 = \cdots = X_N = X$, then*

$$C(X_1, \dots, X_N) = H(X).$$

If instead the variables are independent with common law P, then

$$C(X_1, \dots, X_N) = NH(P).$$

Thus unlimited replication of one random outcome and generation of independent outcomes have fundamentally different seed costs.

Approximate sharing. Suppose a deterministic map g predicts $Y = (X_2, \ldots, X_N)$ from X_1 with error probability ε. If Y takes values in an alphabet of size $M \geq 2$, then

$$0 \leq C(X_1, \ldots, X_N) - C(X_1) \leq h_2(\varepsilon) + \varepsilon \log_2(M-1),$$

where h_2 denotes binary entropy.

Indeed, with $E = \mathbf{1}_{\{Y \neq g(X_1)\}}$,

$$H(Y \mid X_1) = H(E \mid X_1) + H(Y \mid E, X_1).$$

The first term is at most $h_2(\varepsilon)$. The second is zero on success and at most $\log_2(M-1)$ on failure, proving the bound.

Boundary of the result. A collective agreement may change which joint law is requested. It cannot reduce the exact seed cost of a fixed joint law below its entropy. In particular, replacing independent outputs by identical copies changes the specification; it is not an exact simulation of the original independent outputs.

Connection to causal simulation. This theorem concerns one fixed joint law. A causal shared seed must satisfy a family of adaptive transcript laws using a decoder consistent at shared histories. The coordinate-seed construction above does not establish that stronger requirement.

Research question. For a specified causal system, characterize when adding another observer preserves the optimal shared-seed entropy, while keeping all prescribed joint transcript laws and local information constraints unchanged. A new theorem would require solving this additional compatibility problem, not merely renaming the entropy identity proved above.